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Final Project Presentation

Industrial Applications of Computer Vision AA. 2025-26

Smart PPE Compliance Monitoring System

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Presented By:

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Problem and Objective

Problem

Industrial workers can be exposed to hazards when required PPE is missing or replaced by non-compliant headwear. The system aims to detect these visible PPE conditions automatically.

Objective

Build an object-detection system that identifies five PPE-related classes and gives visual and alert-based feedback during live video processing.

Final classes

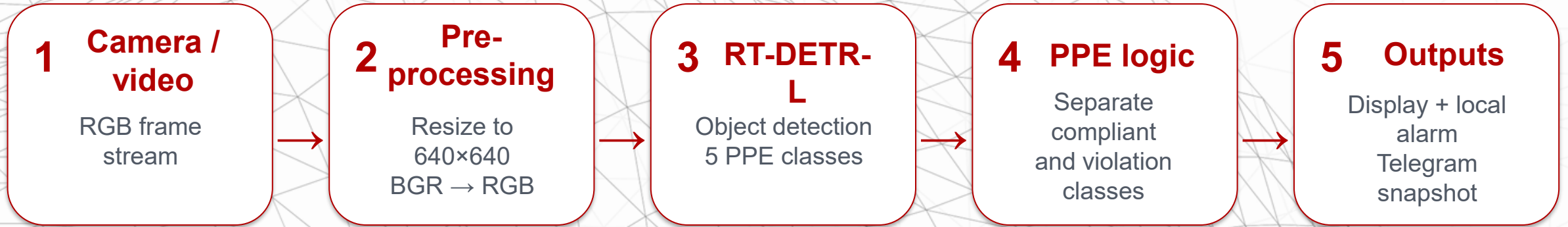
- Bare_Head
- No_Safety_vest
- Non_Compliant_Hat
- Safety_vest
- helmet

Violation classes: Bare_Head, No_Safety_vest, Non_Compliant_Hat

Compliant classes: Safety_vest, helmet



System Pipeline



Implementation notes

- OpenCV handles camera/video capture and display.
- Inference and display are separated using Python threading and frame buffering.
- Violation alerts use the local alarm.wav file and a 5-second Telegram notification cooldown.
- Telegram integration was implemented; end-to-end alerts were simulated and recorded in the terminal rather than treated as a fully validated deployment.



Dataset and Ground Truth

Dataset	Annotated instances	Main purpose
Training	2,085	Model training
Retraining	1,488	Balanced / focused retraining
Evaluation	202	Independent final evaluation

Evaluation dataset

Created from scratch from multiple YouTube videos to represent realistic industrial scenes. It was kept separate from the training/retraining data.

Evaluation class distribution

Class	Instances
Bare_Head	15
No_Safety_vest	65
Non_Compliant_Hat	17
Safety_vest	37
helmet	68

Annotation quality control

After the initial annotation process, every annotated instance was personally inspected. Incorrect bounding boxes, class assignments, and missing/incorrect annotations were corrected before training or evaluation.



Model and Training Setup

Model

RT-DETR-L
PyTorch weights (.pt) used as the final model format.

Final input

600 × 600 pixels
RGB preprocessing
Visible-light RGB video

Final training

30 epochs
batch 4
Optimizer: auto
pretrained initialization

Key final configuration

- Dataset: datasets/retrain_03
- Learning rate: lr0 = 0.01, lrf = 0.01
- Warm-up: 3 epochs
- Scale: 0.8; horizontal flip: 0.5
- AMP enabled; deterministic seed = 0

Important: the four runs changed more than one factor between stages, so they are presented as iterative configuration experiments, not one-variable causal ablations.



Evaluation Methodology

Ground truth

202 annotated instances across five classes. All instances of each class was personally annotated and inspected before evaluation.

Metrics

Precision, recall, mAP@0.50 and mAP@0.50:0.95. Recall is particularly important because missed PPE violations are safety-relevant.

Baseline

The initial RT-DETR-L training configuration provides the reference point for measuring the effect of subsequent trainings changes.

Proposal targets

Metric	Target
FPS	≥ 25 FPS
mAP@0.50	≥ 0.80
Precision	≥ 0.75
Recall	Maximize

Evaluation was performed against the independent evaluation dataset rather than the training data.



Iterative Training Experiments

Run	Main changes	Precision	Recall	mAP50	mAP50-95
Baseline	30 ep.; 640 px; batch 2	0.8933	0.7713	0.7500	0.3946
Retraining 1	40 ep.; 800 px; batch 2	0.8117	0.8036	0.7591	0.3710
Retraining 2	Balanced data; 80 ep.; 600 px; batch 4	0.9126	0.7319	0.7130	0.4011
Final	Retrain-03 data; 30 ep.; 600 px	0.9294	0.8618	0.8542	0.4522

What the results showed

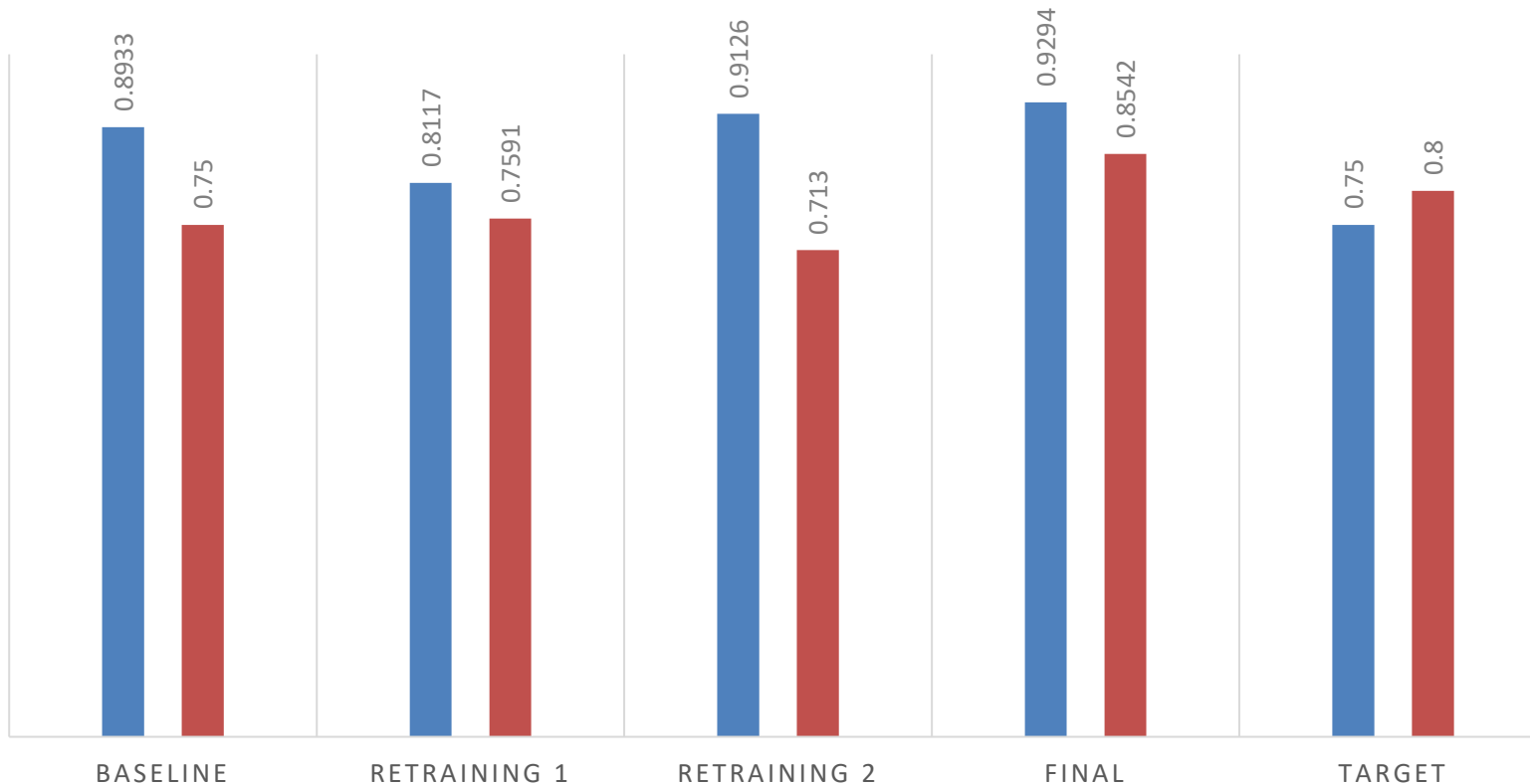
- Baseline mAP@0.50 = 0.7500, below the 0.80 proposal target → further training was needed.
- Retraining 1 slightly increased mAP@0.50 and recall, but mAP@0.50:0.95 decreased and Non_Compliant_Hat performance fell.
- Retraining 2 increased precision and mAP@0.50:0.95 relative to Retraining 1, but mAP@0.50 and recall decreased.
- The final configuration produced the strongest overall results across the four tested runs.



Experimental Results

PERFORMANCE ACROSS THE FOUR TESTED CONFIGURATIONS

■ Precision ■ mAP@0.50



Final mAP@0.50

0.8542

Target: 0.80

Final precision

0.9294

Target: 0.75

Final recall

0.8618

Highest among the four runs

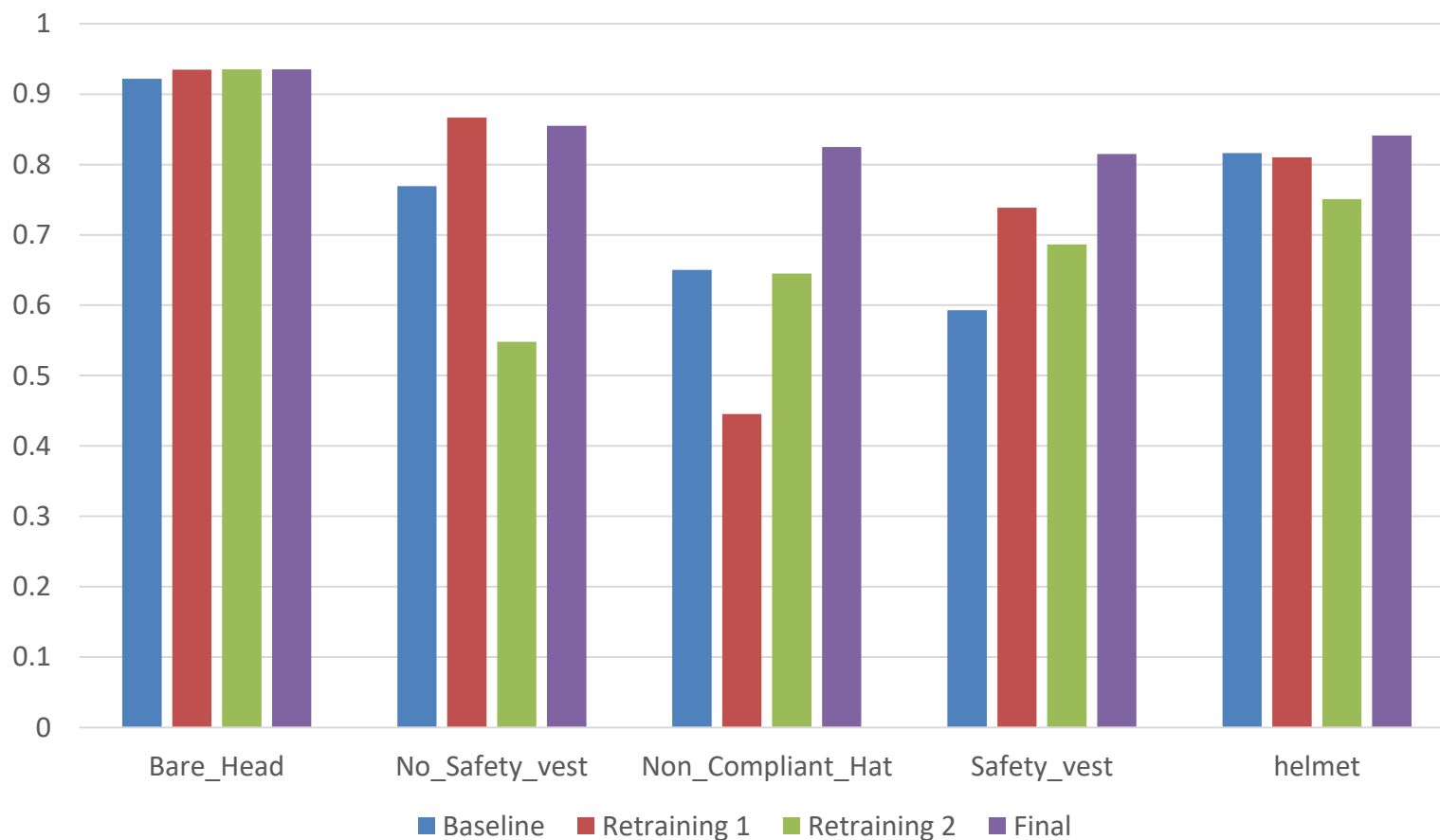
Final mAP@0.50:0.95

mAP@0.50:0.95 also reached its best value in the final run: 0.4522.



Per-Class Performance

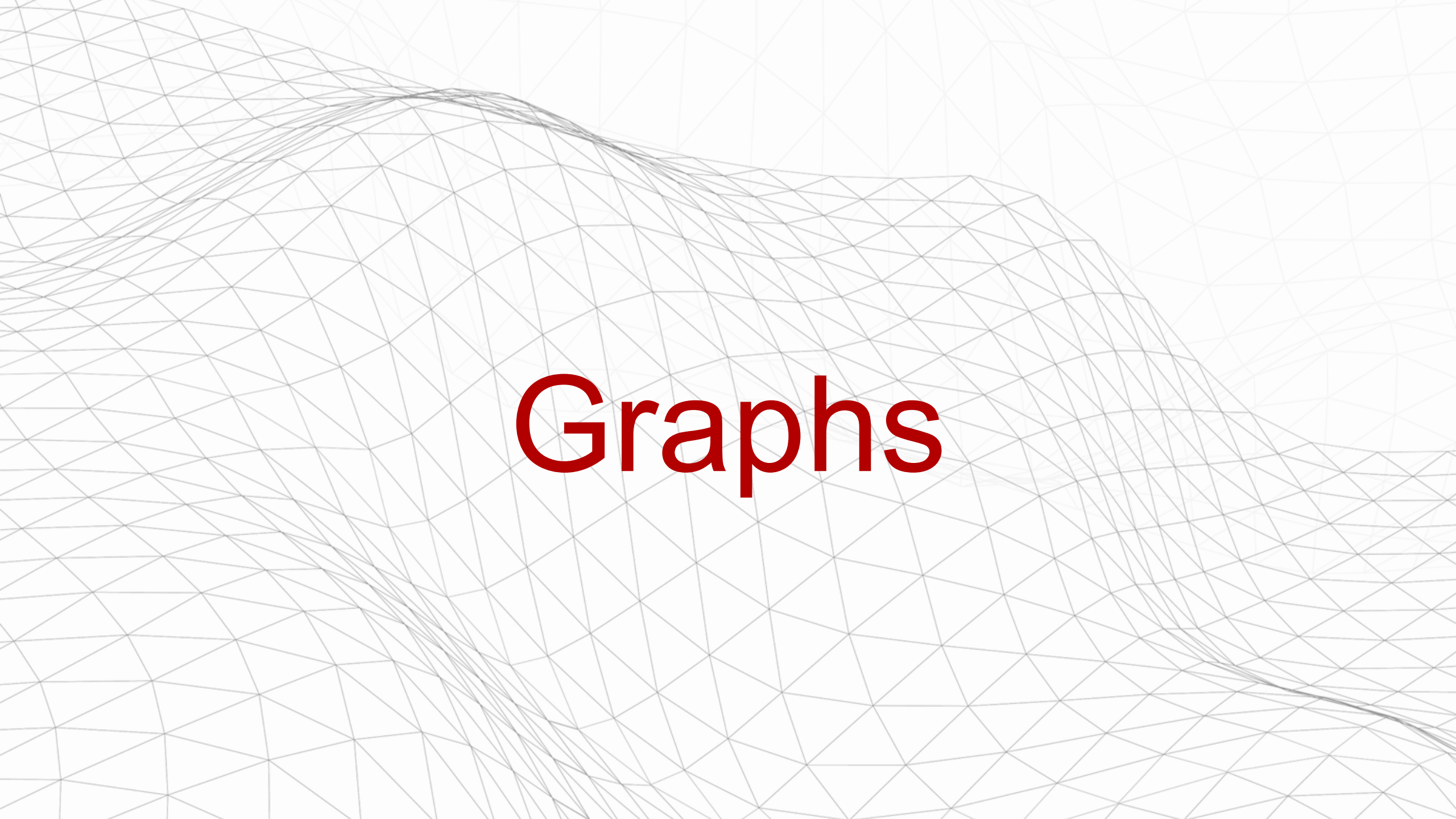
Final mAP@0.50 compared with the baseline



Class	Baseline	Final	Δ
Bare_Head	0.9217	0.9350	+0.0133
No_Safety_vest	0.7693	0.8551	+0.0858
Non_Compliant_Hat	0.6503	0.8250	+0.1747
Safety_vest	0.5928	0.8151	+0.2223
helmet	0.8162	0.8410	+0.0248

Largest gains

- Safety_vest: +0.2223
- Non_Compliant_Hat: +0.1747
- No_Safety_vest: +0.0858

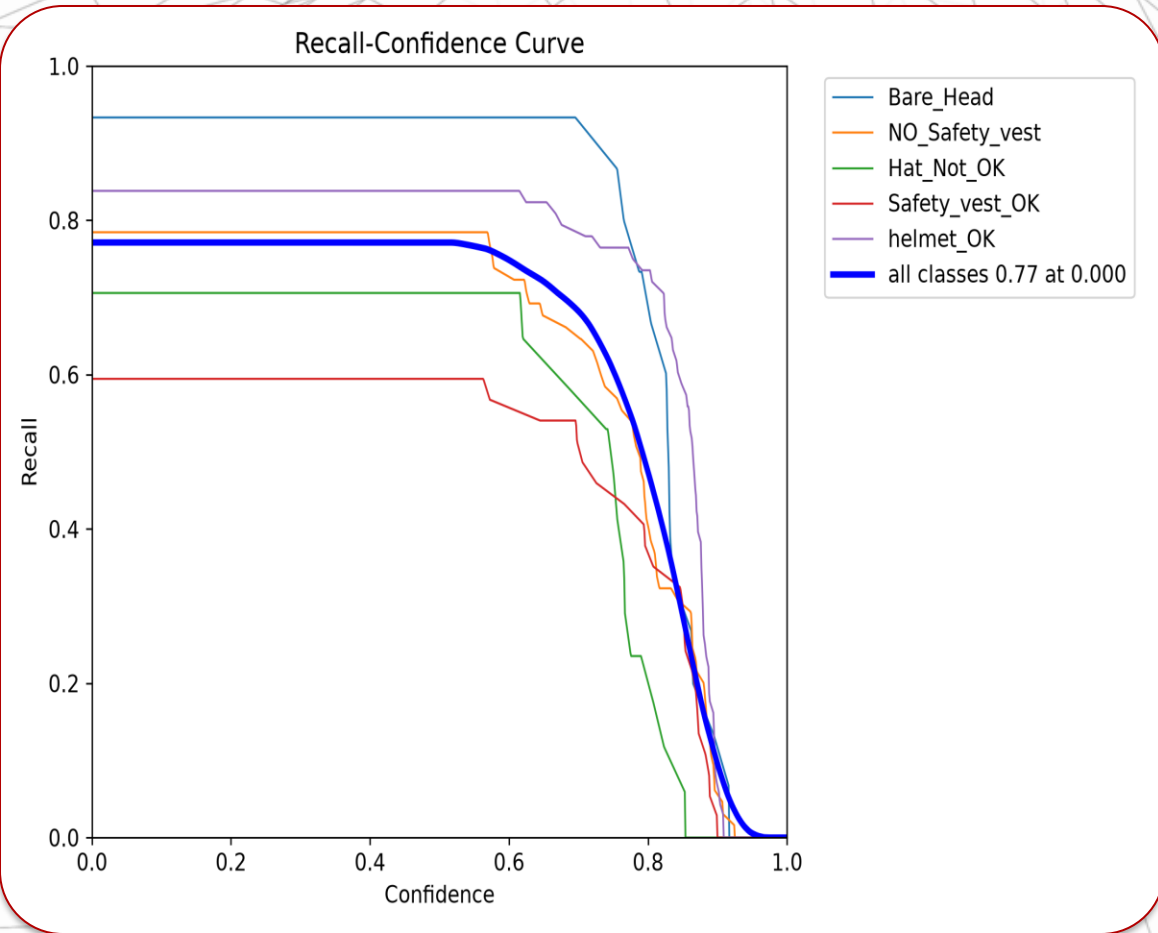
The background features a light gray wireframe mesh that forms a wavy, undulating surface, resembling a topographical map or a stylized landscape. The mesh is composed of numerous small, interconnected triangles. The word "Graphs" is centered in a bold, red, sans-serif font.

Graphs

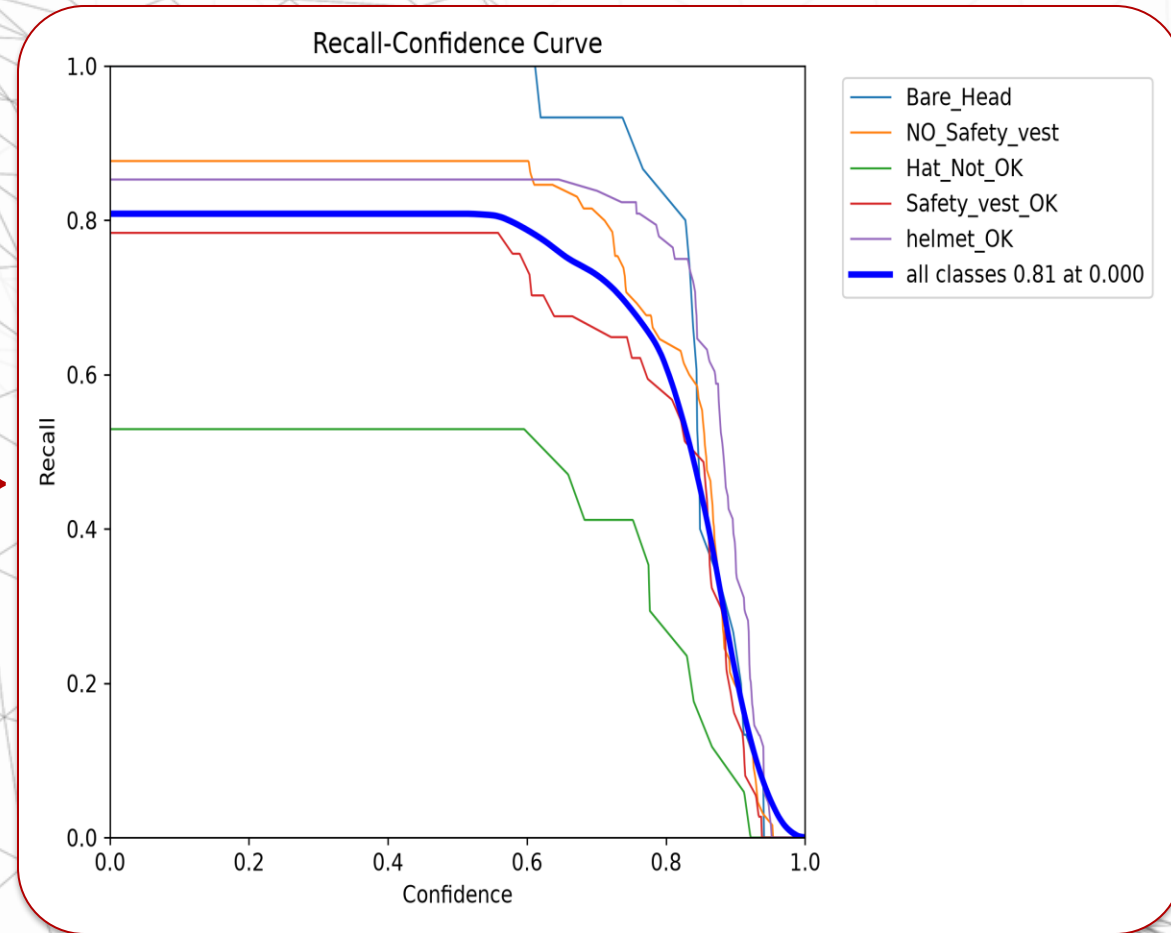


Recall – Confidence Curve of Custom Evaluation Dataset

Baseline/ Initial Training



Retraining - 1

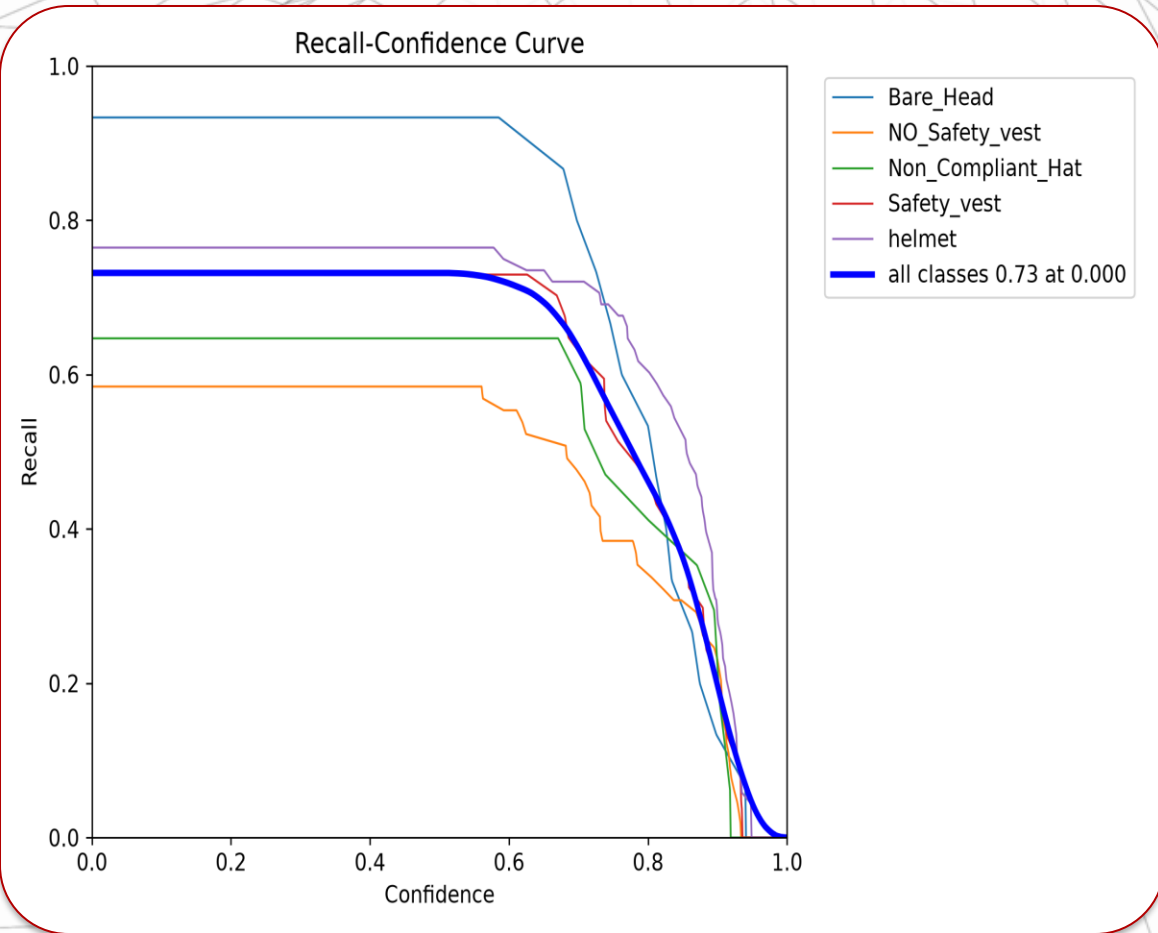


Class names were changed, in order to simplify Hat_Not_OK to Non_Compliant_Hat, Safety_Vest_ok to Safety_Vest, Helmet_OK to Helmet

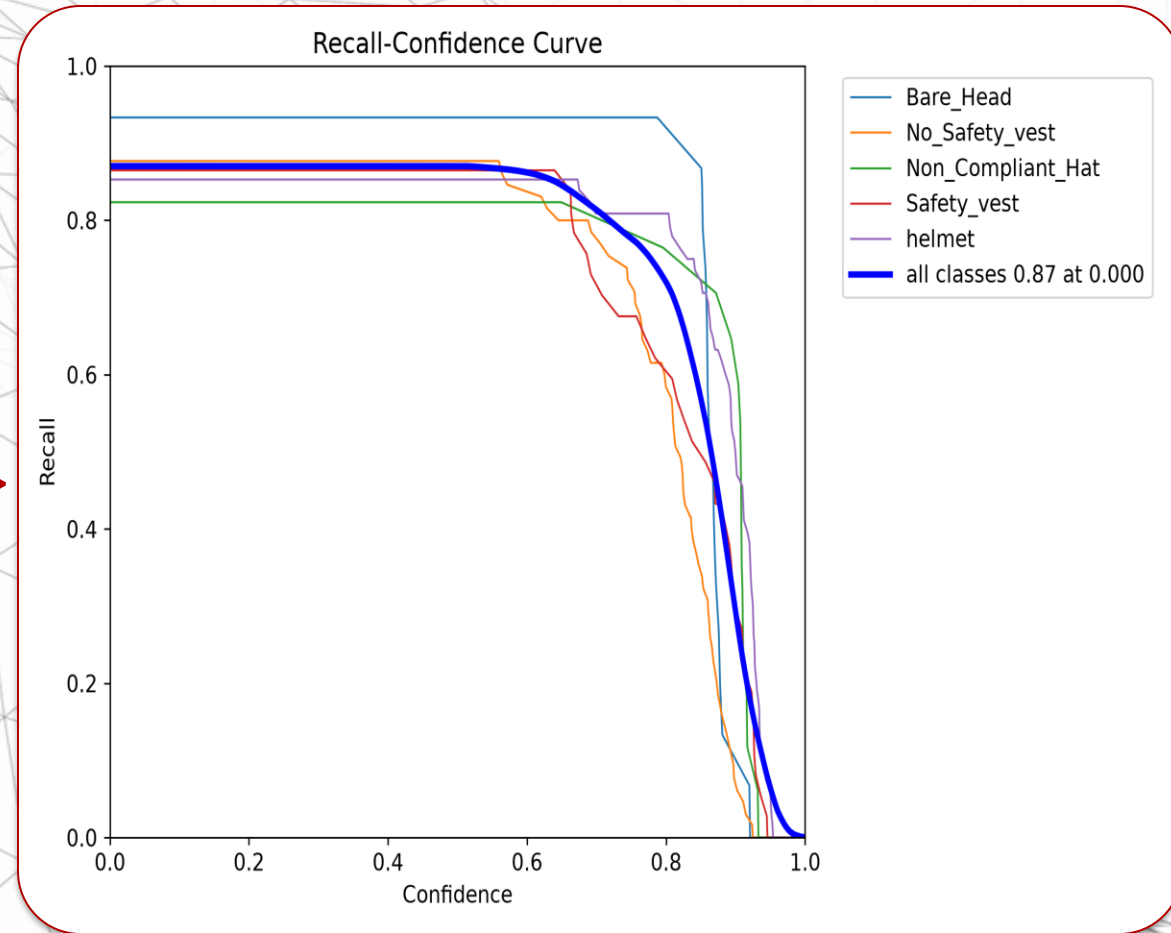


Recall – Confidence Curve of Custom Dataset Evaluation

Retraining - 2



Final Model

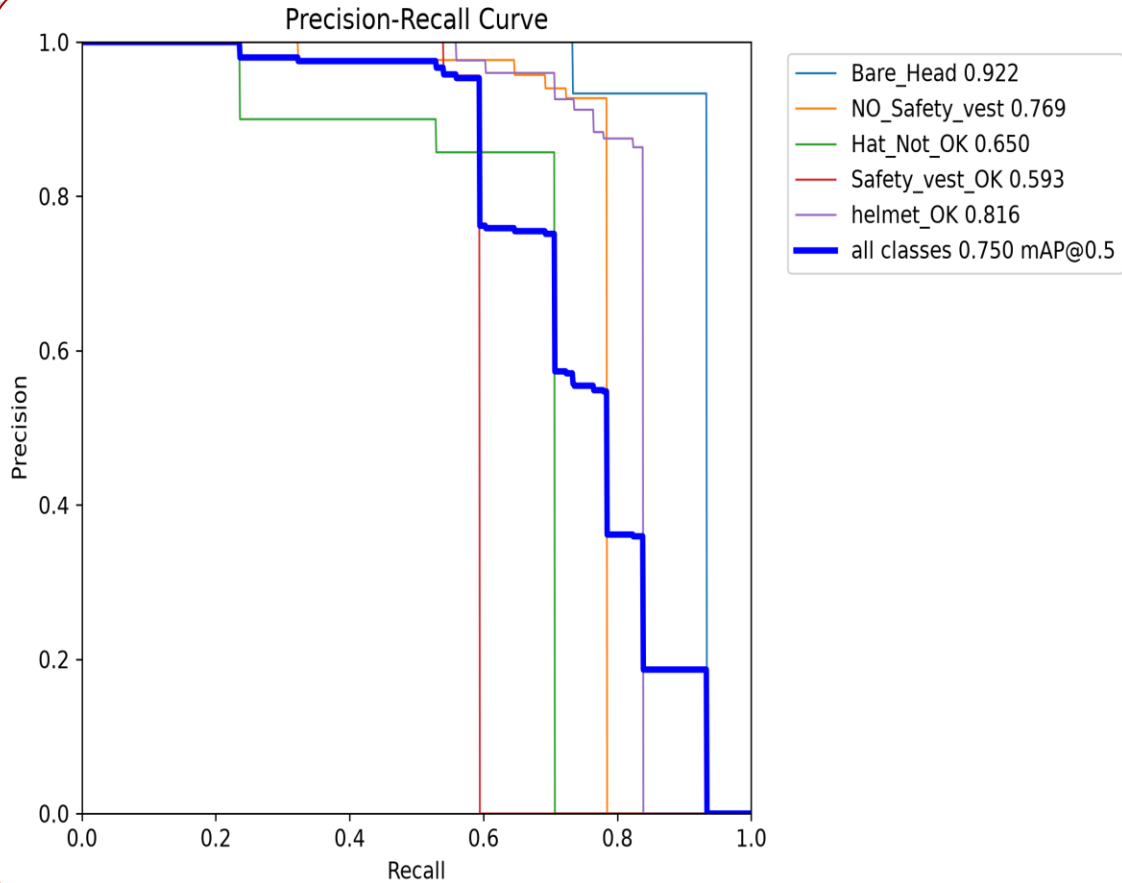


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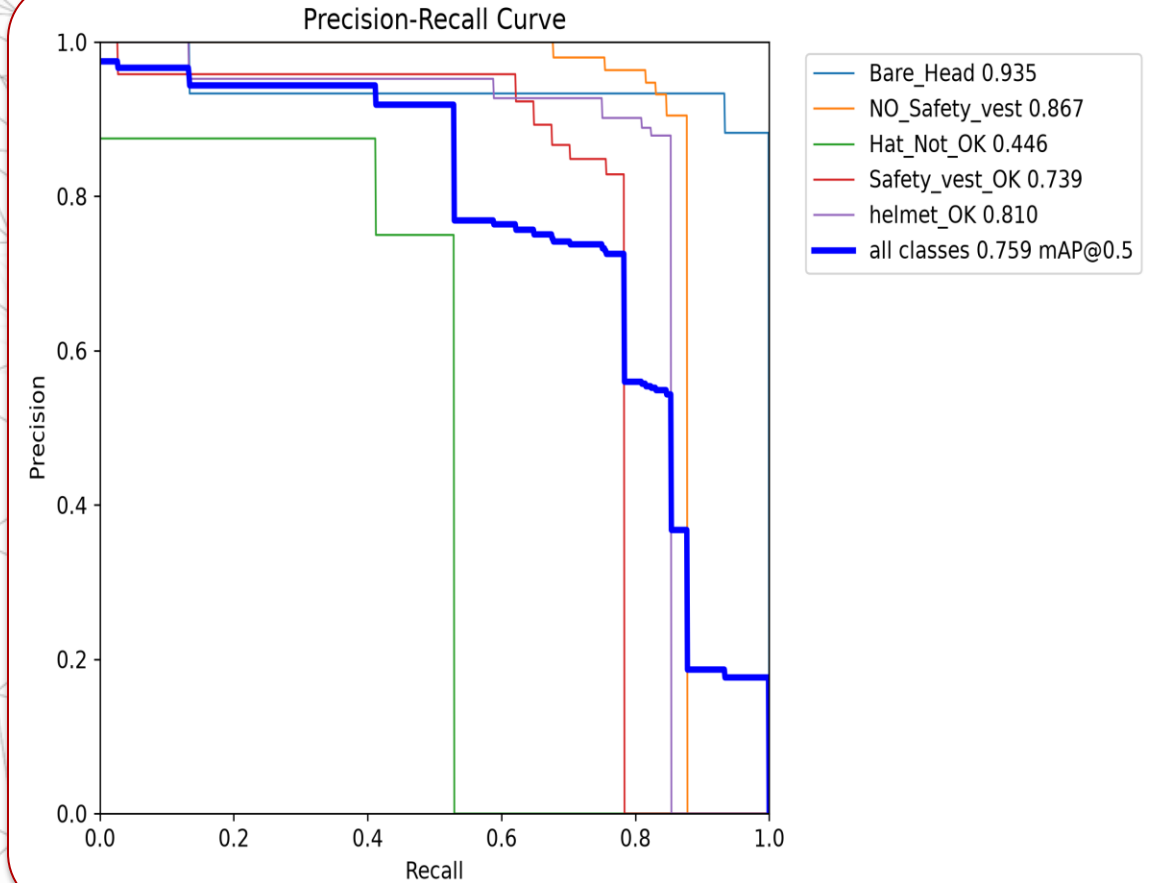


Precision - Recall Curve of Custom Dataset Evaluation

Baseline/ Initial Training



Retraining - 1

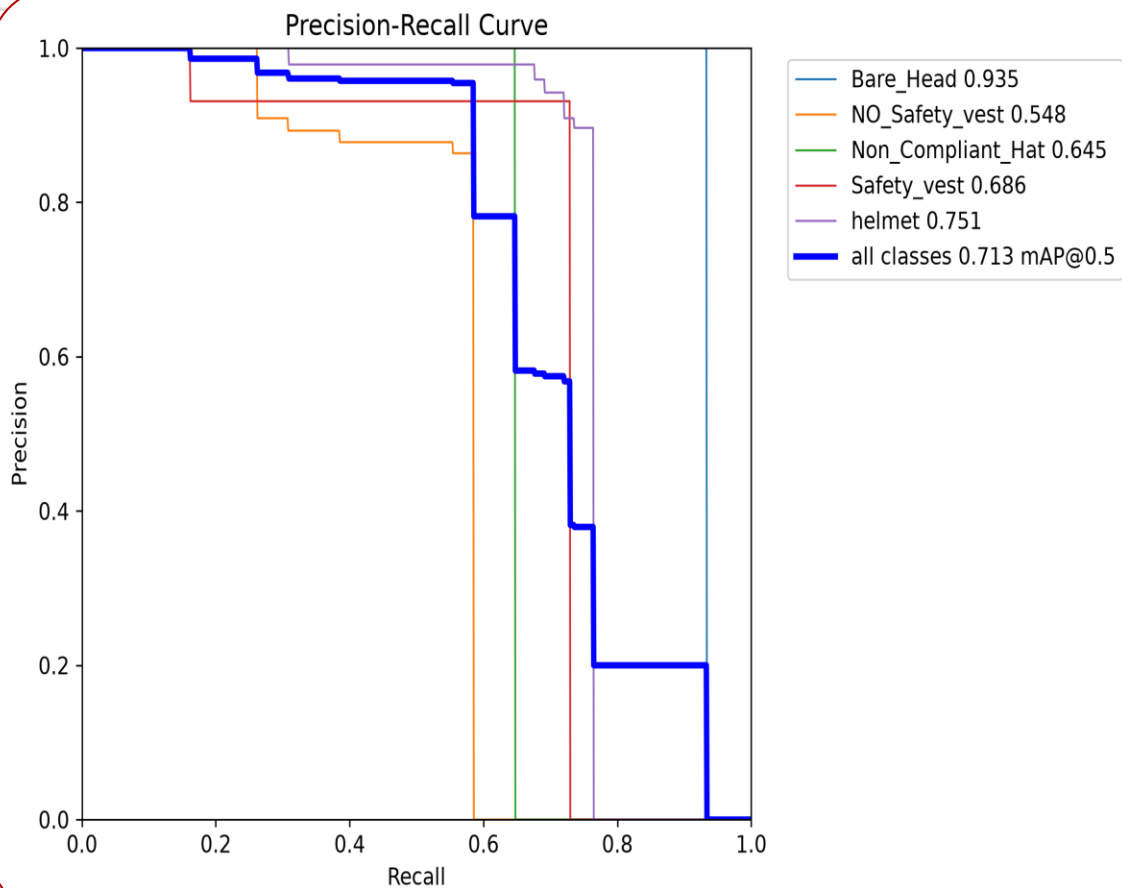


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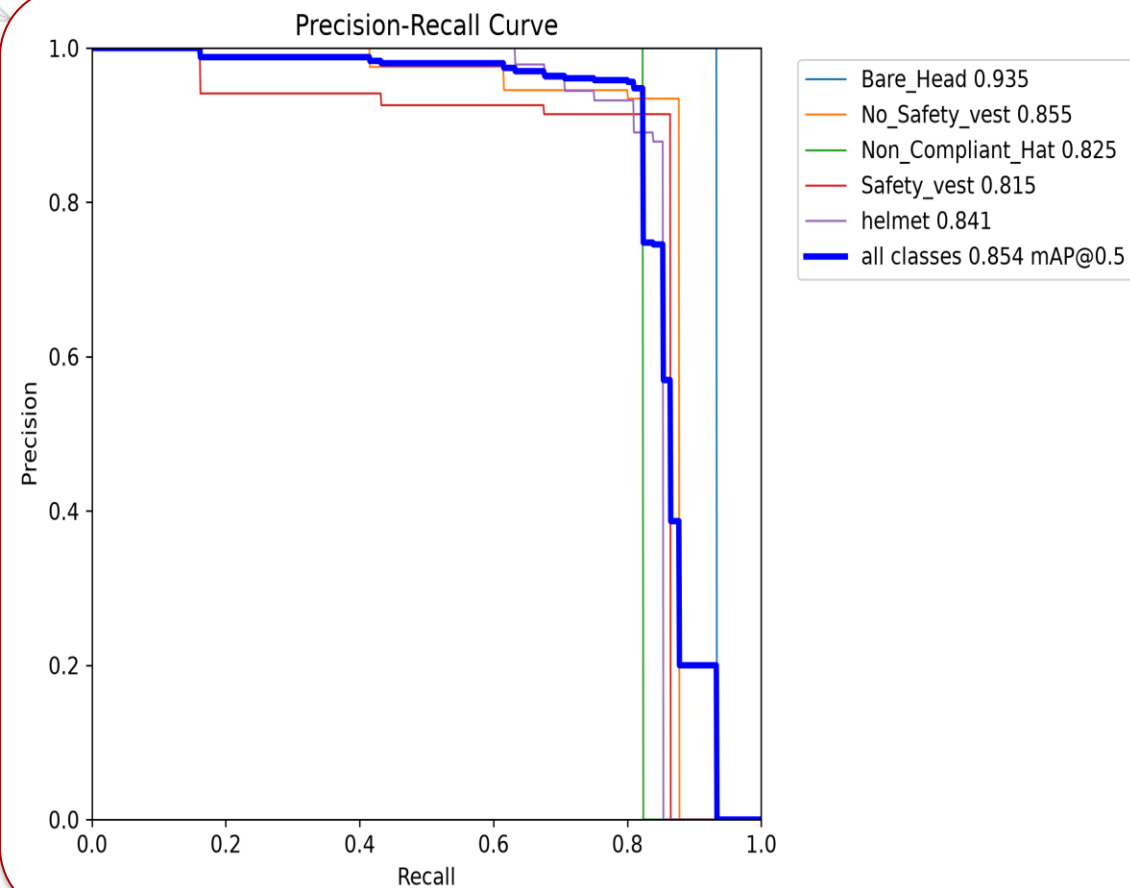


Precision - Recall Curve of Custom Dataset Evaluation

Retraining - 2



Final Model



Class names were changed, in order to simplify Hat_Not_OK to Non_Compliant_Hat, Safety_Vest_ok to Safety_Vest, Helmet_OK to Helmet



Baseline vs Final and Proposal Targets

Metric	Baseline	Final	Change	Target
Precision	0.8933	0.9294	+0.0361	≥ 0.75
Recall	0.7713	0.8618	+0.0905	Maximize
mAP@0.50	0.7500	0.8542	+0.1042	≥ 0.80
mAP@0.50:0.95	0.3946	0.4522	+0.0576	—
Inference FPS	—	~6.5	—	≥ 25

Performance Targets

mAP@0.50: PASSED
Precision: PASSED

Recall

Improved from 0.7713 to 0.8618

Speed target

NOT MET
~6.5 FPS inference vs ≥ 25 FPS



Configuration Analysis and Limitations

Configuration changes across stages

Transition	Important changes
Baseline → R1	640→800 px; 30→40 epochs
R1 → R2	Balanced dataset; 800→600 px; 40→80 epochs; batch 2→4; patience 100→20; other augmentation/settings changed
R2 → Final	New retraining dataset; 80→30 epochs; patience 20→100; HSV/scale/worker settings changed

Interpretation

The experiments demonstrate the performance of tested configurations. Because several parameters and dataset versions changed together, the results do not prove that one individual parameter caused the final improvement.

Remaining limitations

- Inference speed is below the original ≥ 25 FPS target.
- Evaluation data are representative of the collected YouTube scenarios, not a full industrial deployment.
- Telegram notification functionality was implemented but not validated as an end-to-end production service.



Final system and reproducibility

Component	Role
src/	Codes of Training, Retraining and Production
weights/	Final trained PyTorch .pt weights
assets/	Sample images, alarm.wav
docs/	Final presentation
README.md	Setup, dataset links, experiments, references

Run flow

Install Python dependencies → place model weights → configure input source → run production_pipeline.py

External data

Large datasets are kept outside GitHub and linked from the README.

Safety scope

Visible-light PPE detection only. No darkness/IR, ReID, or multi-camera tracking.

The background features a light gray wireframe mesh that forms a series of overlapping, wavy, and undulating shapes, creating a sense of depth and movement. The lines are thin and gray, set against a white background.

Samples



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Compliant Worker



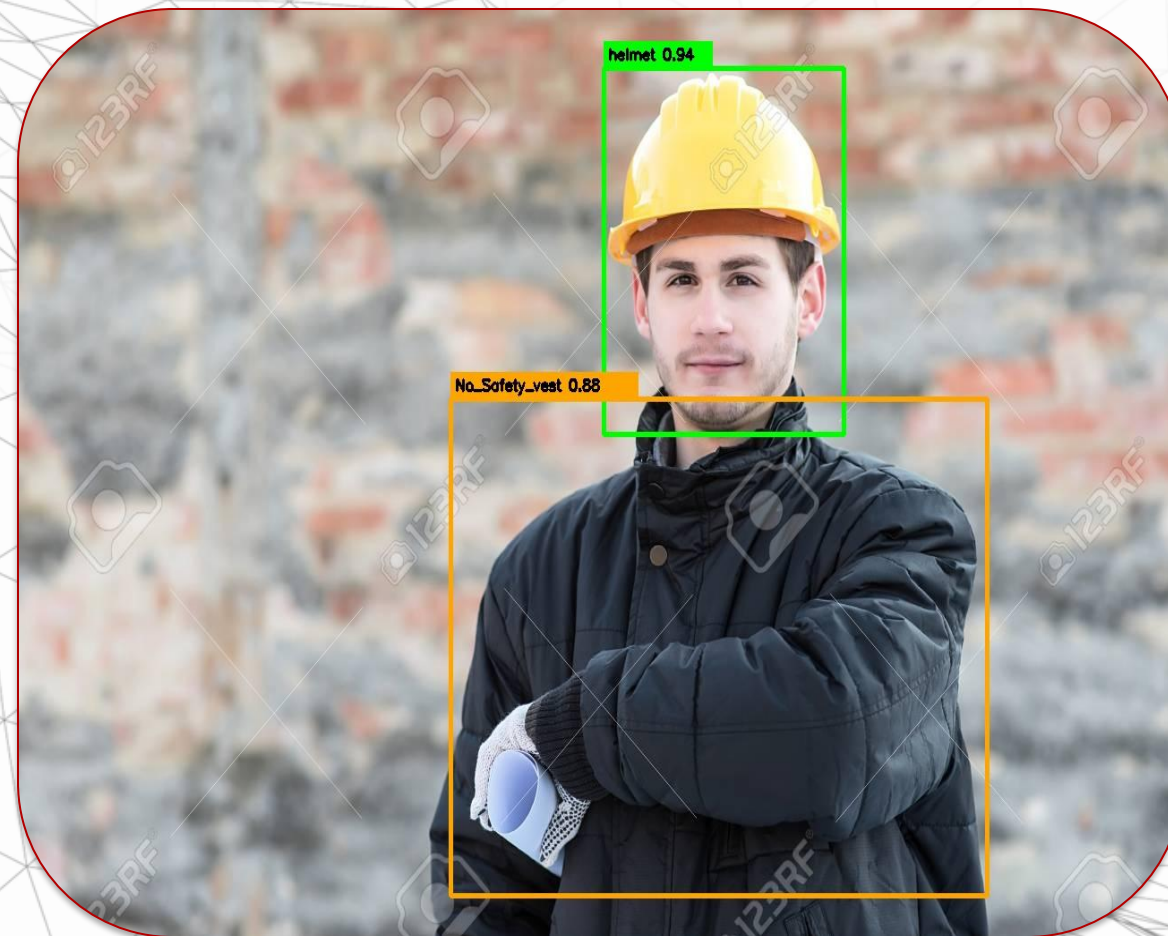


Violation – Bare Head With Compliant Safety Vest





Violation – No Safety Vest with Compliant Helmet





Violation – Non-Compliant Hat with Compliant Safety Vest



The background features a light gray wireframe mesh that forms a series of overlapping, wavy, and undulating shapes, creating a sense of depth and movement. The lines are thin and gray, set against a plain white background.

Thank You